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INSPECTION OF LITHOGRAPHIC LAYERS AND DYNAMIC DATA VIA INLINE METROLOGY

Abstract

Aspects of disclosure provide a method for processing an apparatus. The method includes determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate. The method includes depositing a photoresist over the substrate and developing the photoresist to form one or more developed regions and one or more undeveloped regions. One of the one or more developed regions or the one or more undeveloped regions of the photoresist are removed to form an altered electrical connection template. The altered component and the altered electrical connection template are scanned to identify one or more misaligned regions of the altered electrical connection template. At least a portion of the altered electrical connections template is removed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit of U.S. provisional patent application Ser. No. 63/562,585, filed Mar. 7, 2024, which is herein incorporated by reference.

BACKGROUND

Field

[0002] Embodiments of the present disclosure generally relate to apparatuses, systems, and methods for processing one or more substrates. More specifically, the present disclosure relates adaptive packaging methods and apparatuses.

Description of the Related Art

[0003] Photolithography is widely used in the manufacturing of semiconductor devices and display devices, such as liquid crystal displays (LCDs). Large area substrates are often utilized in the manufacture of LCDs. LCDs, or flat panels, are commonly used for active matrix displays, such as computers, touch panel devices, personal digital assistants (PDAs), cell phones, television monitors, and the like. Generally, flat panels include a layer of liquid crystal material forming pixels sandwiched between two plates. When power from a power supply is applied across the liquid crystal material, an amount of light passing through the liquid crystal material is controlled at pixel locations, enabling images to be generated. Other manufacturing techniques are used to produce organic light emitting diode (“OLED”) displays for use in computers, monitors and other systems providing visual output.

[0004] Microlithography techniques have been employed to create electrical features incorporated as part of the liquid crystal material layer forming the pixels. According to these techniques, a light-sensitive photoresist is applied to at least one surface of the substrate. Then, a pattern generator exposes selected areas of the light-sensitive photoresist as part of a pattern with light to cause chemical changes to the photoresist in the selective areas to prepare these selective areas for subsequent material removal and/or material addition processes to create the electrical features.

[0005] In order to continue to provide display devices and other devices at the prices demanded by consumers, new apparatuses and approaches are needed to precisely and cost-effectively create patterns on substrates, such as large area substrates.

[0006] In digital lithography tools, images from a camera are used to find a position of alignment marks so that processing may occur at a known location. In order to obtain the images, cameras are calibrated and specially chosen for pixel size, orientation (rotation), and uniformity.

[0007] One of the major challenges of microlithography systems is placement of wiring between components. Often, arrangements of components on a substrate involve a defined alignment, location, and orientation. Wiring must be placed from one component to another component on substrate. Placement of the component within on the substrate is critical. Generally, the wired connection from one component to another component is fixed (static). If one component is misaligned regarding its intended position compared to another component, then the wiring will be misaligned when created. However, misalignment can occur during processing. The problems that this causes range from components produced that are out of specification to components that do not work altogether. Problems of these sort impact the overall economic viability of the lithography process.

[0008] There is a need to enable the rapid processing, high throughput creation of wiring systems for components such that the components may be misaligned to a threshold amount during

processing. Further, there is a need to enable the slight deviations in the placement of components in a defined field such that during processing, systems may adapt to the altered processing of components “in-situ” rather than a hypothetical perfect alignment that is difficult to achieve.

SUMMARY

[0009] Aspects of disclosure provide a method for attaching wired connections between components using both design and field measured data of the components to produce accurate wired connections.

[0010] In one example embodiment, a method for processing an apparatus in a lithography system is disclosed. The method includes determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate. The method includes depositing a photoresist over the substrate and developing the photoresist to form one or more developed regions and one or more undeveloped regions. One of the one or more developed regions or the one or more undeveloped regions of the photoresist are removed to form an altered electrical connection template. The altered component and the altered electrical connection template are scanned to identify one or more misaligned regions of the altered electrical connection template. At least a portion of the altered electrical connections template is removed.

[0011] In another example embodiment, a method for processing an apparatus in a microlithography system is disclosed. The method includes determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate. A photoresist development process is performed. The photoresist development process includes depositing a photoresist over the substrate and developing the photoresist to form one or more developed regions and one or more undeveloped regions. The one or more developed regions or the one or more undeveloped regions of the photoresist are removed to form an altered electrical connection template. The altered component and the altered electrical connection template are scanned to identify one or more misaligned regions of the altered electrical connection template. At least a portion of the altered electrical connection template are removed upon identifying the one or more misaligned regions. The photoresist development process is repeated upon identifying the one or more misaligned regions. The one or more altered electrical connections are formed.

[0012] In one example embodiment, a method for processing a substrate in a microlithography system is disclosed. The method includes determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate and performing a photoresist development process. The photoresist development process includes depositing a photoresist over the substrate and developing the photoresist to form one or more developed regions and one or more undeveloped regions. The one or more developed regions or the one or more undeveloped regions of the photoresist are removed to form an altered electrical connection template. The altered component and the altered electrical connection template are scanned to identify one or more misaligned regions of the altered electrical connection template. At least a portion of the altered electrical connection template are removed upon identifying the one or more misaligned regions. The photoresist development process is repeated upon identifying one or more misaligned regions. An electrical connection material deposited in the altered electrical connection template. At least a portion of the altered electrical connections template are removed to form an altered layout plot.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above,

may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

[0014] FIG. 1A is a perspective view of a photolithography system, according to embodiments.

[0015] FIG. 1B is a perspective view of a photolithography system having two stages, according to embodiments.

[0016] FIG. 2A is a perspective schematic view of an image projection apparatus, according to embodiments.

[0017] FIG. 2B is a perspective schematic view of an image projection apparatus including one or more micro-LEDs, according to embodiments.

[0018] FIG. 2C is a perspective schematic view of an image projection apparatus including one or more digital micro-mirror devices (DMDs), according to embodiments.

[0019] FIG. 3 is a schematic depiction of a designed layout plot, according to embodiments.

[0020] FIG. 4 is a schematic depiction an altered layout plot, according to embodiments.

[0021] FIG. 5 is a material addition method for processing an apparatus in a lithography system, according to embodiments.

[0022] FIG. 6A-6F are schematic, top views of a substrate during the method of FIG. 5 for processing an apparatus in a lithography system, according to embodiments.

[0023] FIG. 7 is a material subtraction method for processing an apparatus in a lithography system, according to embodiments.

[0024] FIG. 8A-8G are schematic, top views of a substrate during a method of FIG. 7 for processing an apparatus in a lithography system, according to embodiments.

[0025] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0026] Embodiments of the present disclosure generally relate to apparatuses, systems, and methods for processing one or more substrates. More specifically, the present disclosure relates to adaptive packaging methods and apparatuses.

[0027] FIG. 1A is a perspective view of a photolithography system **100A** according to embodiments disclosed herein. The photolithography system **100A** includes a base frame **110**, a slab **120**, a stage **130**, and a processing apparatus **160**. The base frame **110** rests on the floor of a fabrication facility and supports the slab **120**. Passive air isolators **112** are positioned between the base frame **110** and the slab **120**. In one embodiment, the slab **120** is a monolithic piece of granite, and the stage **130** is disposed on the slab **120**. A substrate **140** is supported by the stage **130**. A plurality of holes are formed in the stage **130** for allowing a plurality of lift pins to extend therethrough. In some embodiments, the lift pins rise to an extended position to receive the substrate **140**, such as from one or more transfer robots (not shown). The one or more transfer robots are used to load and unload a substrate **140** from the stage **130**.

[0028] The substrate **140** comprises any suitable material, for example, quartz used as part of a flat panel display. In other embodiments, the substrate **140** is made of other materials. In some embodiments, the substrate **140** has a photoresist layer formed thereon. A photoresist is sensitive to radiation. A positive photoresist includes portions of the photoresist which, when exposed to radiation, will be respectively soluble to photoresist developer applied to the photoresist after the pattern is written into the photoresist. A negative photoresist includes portions of the photoresist which, when exposed to radiation, will be respectively insoluble to photoresist developer applied to the photoresist after the pattern is written into the photoresist. The chemical composition of the photoresist determines whether the photoresist will be a positive photoresist or negative

photoresist. Examples of photoresists include, but are not limited to, at least one of diazonaphthoquinone, a phenol formaldehyde resin, poly(methyl methacrylate), poly(methyl glutarimide), and SU-8. In this manner, the pattern is created on a surface of the substrate **140** to form the electronic circuitry.

[0029] The photolithography system **100A** includes a pair of supports **122** and a pair of tracks **124**. The pair of supports **122** are disposed on the slab **120**, and the slab **120** and the pair of supports **122** are a single piece of material. The pair of tracks **124** is supported by the pair of the supports **122**, and the stage **130** moves along the tracks **124** in the X-direction. In one embodiment, the pair of tracks **124** is a pair of parallel magnetic channels. As shown, each track **124** of the pair of tracks **124** is linear. In other embodiments, one or more tracks **124** are non-linear. An encoder **126** is coupled to the stage **130** in order to provide location information to a controller.

[0030] The processing apparatus **160** includes a support **162** and a processing unit **164**. The support **162** is disposed on the slab **120** and includes an opening **166** for the stage **130** to pass under the processing unit **164**. The processing unit **164** is supported by the support **162**. In one embodiment, the processing unit **164** is a pattern generator configured to expose a photoresist in a photolithography process. In some embodiments, the pattern generator is configured to perform a maskless lithography process. The processing unit **164** includes a plurality of image projection apparatus (shown in FIGS. 2A and 2B). In one embodiment, the processing unit **164** contains as many as **84** image projection apparatuses. Each image projection apparatus is disposed in a case **165**. The processing apparatus **160** is useful to perform maskless direct patterning.

[0031] During operation, the stage **130** moves in the X-direction from a loading position, as shown in FIG. 1A, to a processing position. The processing position is one or more positions of the stage **130** as the stage **130** passes under the processing unit **164**. During operation, the stage **130** is lifted by a plurality of air bearings and moves along the pair of tracks **124** from the loading position to the processing position. A plurality of vertical guide air bearings are coupled to the stage **130** and positioned adjacent an inner wall **128** of each support **122** in order to stabilize the movement of the stage **130**. The stage **130** also moves in the Y-direction by moving along a track **150** for processing and/or indexing the substrate **140**. The stage **130** is capable of independent operation and can scan a substrate **140** in one direction and step in the other direction.

[0032] A metrology system measures the X and Y lateral position coordinates of each of the stage **130** in real time so that each of the plurality of image projection apparatus can accurately locate the patterns being written in a photoresist covered substrate. The metrology system also provides a real-time measurement of the angular position of each of the stage **130** about the vertical or Z-axis. The angular position measurement can be used to hold the angular position constant during scanning by means of a servo mechanism, or the angular position measurement can be used to apply corrections to the positions of the patterns being written on the substrate **140** by the image projection apparatus **270**, shown in FIGS. 2A-2B. These techniques may be used in combination.

[0033] FIG. 1B is a perspective view of a photolithography system **100B** according to embodiments disclosed herein. The system **100B** is similar to the system **100**; however, the system **100B** includes two stages **130**. Each of the two stages **130** is capable of independent operation and can scan a substrate **140** in one direction and step in the other direction. In some embodiments, when one of the two stages **130** is scanning a substrate **140**, another of the two stages **130** is unloading an exposed substrate and loading the next substrate to be exposed.

[0034] While FIGS. 1A-1B depict two embodiments of a photolithography system, other systems and configurations are also contemplated herein. For example, photolithography systems including any suitable number of stages are also contemplated.

[0035] FIG. 2A is a perspective schematic view of an image projection apparatus **270** according to one embodiment, which is useful for a photolithography system, such as system **100A** or system **100B**. The image projection apparatus **270** includes one or more spatial light modulators **280**, an alignment and inspection system **284** including a focus sensor **283** and a camera **285**, and

projection optics **286**. The components of image projection apparatus vary depending on the spatial light modulator being used. Spatial light modulators include, but are not limited to, micro-LEDs, digital micro-mirror devices (DMDs) and liquid crystal displays (LCDs).

[0036] In operation, the spatial light modulator **280** is used to modulate one or more properties of the light, such as amplitude, phase, or polarization, which is projected through the image projection apparatus **270** and to a substrate, such as the substrate **140**. The alignment and inspection system **284** is used for alignment and inspection of the components of the image projection apparatus **270**. In one embodiment, the focus sensor **283** includes a plurality of lasers which are directed through the lens of the camera **285** and the back through the lens of the camera **285** and imaged onto sensors to detect whether the image projection apparatus **270** is in focus. The camera **285** is used to image the substrate **140** to ensure the alignment of the image projection apparatus **270** and photolithography system **100A** or **100B** is correct or within a predetermined tolerance. The projection optics **286**, such as one or more lenses, is used to project the light onto the substrate, such as the substrate **140**.

[0037] FIG. 2B is an image project apparatus **271** according to embodiments described herein. In the embodiment shown in FIG. 2B, the image projection apparatus **271** includes one or more micro-LEDs **287** as the spatial light modulator(s), a focus sensor and inspection system **284** and projection optics **286**. In one embodiment, the image projection apparatus **271** further includes a beamsplitter. MicroLEDs are microscopic (for example, less than about 100 μm) light emitting diodes, which may be arranged in an array and used to form the individual pixels of a substrate, such as a display device. MicroLEDs include inorganic materials, such as an inorganic Gallium Nitride (GaN) material. Since micro-LEDs are self-emitting, an outside light source is not needed in the image projection apparatus **271**.

[0038] In embodiments using micro-LEDs, the camera **285** is also useful to measure the image pixel pitch of the one or more micro-LEDs to calibrate for any thermal expansion happening at the micro-LED device.

[0039] FIG. 2C is an image projection apparatus **281** according to embodiments described herein. In the embodiment shown in FIG. 2C, the image projection apparatus **281** uses one or more DMDs **289** as the spatial light modulator(s). The image projection apparatus **281** is part of an image projection system **290**, which includes a light source **272**, an aperture **274**, a lens **276**, a frustrated prism assembly **288**, one or more DMDs **289**, and a light dump **282**, in addition to the alignment and inspection system **284** and the projection optics **286**. The light source **272** is any suitable light source, such as a light emitting diode (LED) or a laser, capable of producing a light having a predetermined wavelength. In one embodiment, the predetermined wavelength is in the blue or near ultraviolet (UV) range, such as less than about 450 nm. The frustrated prism assembly **288** includes a plurality of reflective surfaces. The projection optics **286** can include, as an example, a 10 $\times$ objective lens. During operation of the image projection apparatus **281** shown in FIG. 2C, a light beam **273** having a predetermined wavelength, such as a wavelength in the blue range, is produced by the light source **272**. The light beam **273** is reflected to the DMD **289** by the frustrated prism assembly **288**. The DMD includes a plurality of mirrors, and the number of mirrors corresponds to the number of pixels to be projected. The plurality of mirrors are individually controllable, and each mirror of the plurality of mirrors is at an “on” position or “off” position, based on the mask data provided to the DMD **289** by the controller. When the light beam **273** reaches the mirrors of the DMD **289**, the mirrors that are at “on” position reflect the light beam **273**, i.e., forming the plurality of write beams, to the projection optics **286**. The projection optics **286** then projects the write beams to the surface of the substrate **140**. The mirrors that are at “off” position reflect the light beam **273** to the light dump **282** instead of the surface of the substrate **140**.

[0040] In processing of materials through the lithography system, components with one or more dice are placed on a stage for processing. While a designed layout of the components may be achieved for processing, various factors may impact the designed layout. Processing speed of the

lithography system may slightly move the components as the lithography system handles the components. These slight movements, for components that are small, may have drastic impact on the final product as wiring from an origination point to ending point can be small. Each component, furthermore, may have several dice, therefore many wired connections may be impacted by incorrect alignment of the component compared to the designed layout.

[0041] Aspects of the disclosure provide that, given a designed placement of one or more dice within a packaging layout as well as altered placements that may involve some errors, the designed placement may be modified or distorted to match the altered placements, thereby adaptively generating wiring for packaging of dice into a larger assembly. The disclosure further provides for measuring the location and orientation of the adaptively generated wiring to detect defects in the adaptively generated wiring prior to the deposition of the wiring material.

[0042] FIG. 3 is a designed layout plot 300. The designed layout plot 300 includes designed components 301 disposed on the substrate 140. The plurality of designed components 301 are connected using a plurality of designed wired connections 302 from either a fixed perimeter 303 of the substrate 140 to a designed component 301 or from designed component 301 to another designed component 301. The designed components 301 further include an edge 304. The designed wired connections 302 (e.g., designed electrical connections) may be connected to the designed components 301 at the edge 304 of the designed components 301. Such designed layout plot 300 would occur if each of the designed components 301 is placed, as intended, on a substrate 140 for processing. As will be understood, placement of components with high accuracy may be difficult. To this end, even if the designed component 301 is placed with high accuracy, shifting may occur during processing, as photolithography systems move large substrates at fast speeds.

[0043] FIG. 4 is an altered layout plot 400 of an altered component 401. As can be seen, the altered component 401 is shifted from the position in FIG. 3. An altered wired connection 402 (e.g., an altered electrical connection) to the altered component 401 is therefore required in order for the altered component 401 to operate correctly. To this end an altered wired connection 402 is produced such that the altered wired connection 402 between altered components 401 or between the altered component 401 and the fixed perimeter 403 is properly established. As with the designed components 301, the altered components 401 include altered edges 404. The altered wired connections 402 may be connected to the altered components 401 at the edge 404 of the altered components 401.

[0044] FIG. 5 is a flow chart of a method 500 for processing an apparatus in a lithography system. The method 500 is a material additional method, i.e., the altered wired connections 402 are formed by depositing a wired connection material in a wired connection region. FIGS. 6A-6F are schematic, top views of a substrate 140 during the method 500 for processing an apparatus in a lithography system. In some embodiments, the method 500 for processing the substrate 140 may be performed in a microlithography system, such as photolithography system 100A or photolithography system 100B.

[0045] At operation 501, designed data may be obtained. The designed data (e.g., a first set of data) may include coordinate data on a designed component 301 as well as a designed wired connection 302 pattern to the designed component 301 in a designed state. This designed data may be obtained from manufacturing drawings or a memory of the controller, for example. In one embodiment, position data is obtained on the designed component 301 and the designed wired connection 302 to the designed component 301 for a designed state. The designed component 301 may be any type of component that needs wiring attachment, such as, but not limited by, processors. The designed wired connection 302 may be, for example, from the designed component 301 to a fixed perimeter 303, or to a predetermined location, e.g., the edge 304 of another designed component 301.

[0046] At operation 502, as shown in FIG. 6A, a substrate 140 with a plurality of altered components 401 is provided. The substrate 140 is placed on a stage of a microlithography system. As can be understood, the altered component 401 is positioned within a defined field for

processing. The altered components **401** may vary slightly from the designed component **301** due to various factors, including the ability of locating such designed components **301** accurately during initial processing of the platen (indexing table) before loading of the platen into the microlithography machine. Other factors may also cause the designed component **301** to move, such as handling of the components during microlithography steps.

[0047] At operation **503**, the altered components **401** are scanned in-situ to obtain altered component data (e.g., a second set of data). To achieve an altered wired connection **402** to the altered components **401**, either from the fixed perimeter **403** or in between altered components **401**, the “in-situ” positioning of the altered components **401** on the substrate **140** is scanned with at least one scanning device to be able to ascertain the exact positioning of each altered component **401**. In some embodiments, the stage including the altered component **401** is scanned with the at least one scanning device. The substrate **140**, therefore, is moved into a position in the photolithography system **100A**, **100B** such that the scanning occurs for processing of coordinate data. As will be understood, scanning of the altered components **401** may also be accomplished by a separate process, and the data fed to the photolithography system **100A**, **100B** for use. The altered component data obtained may be stored in the memory of the controller. The altered component data may be compared to the designed data to determine any offsets that may be present in the placement of the components compared to the designed components **301**. An offset may be calculated to allow the photolithography system **100A**, **100B** to understand the exact placement of the altered components **401**. The wired connection pattern from the designed condition may then be altered using data from the offset calculation. Visual image data may also be used to determine the difference that is needed for the wired connection pattern.

[0048] Computer analysis of the data may be performed to determine the required offset. The computer analysis may not only calculate the offset, but also provide the location of the altered wired connection **402** between the altered component **401** and the fixed perimeter **403** or other altered components **401**, therefore speeding processing.

[0049] In some embodiments, hardware architects may determine that slight variations from designed placement are allowable. The deviation for the wired connection and the component may be determined to be within acceptable levels. In those instances, no alteration of the wired connection may be required.

[0050] When deviations from these acceptable levels are found, a warning may be generated to the processors to allow for notification that alterations/modification of the wiring are necessary. In other embodiments, if the location and orientation of the components is outside of a maximum threshold, wired connections cannot be effectively made to the component. A separate warning may be made to the processors that the components are out of tolerance and that even with modification of the wired connection, such wired connections will be compromised.

[0051] As will be understood, designs for wired connections may take into account not only placement accuracy of the starting and ending points of the wired connection, but also wired connection length. If a length of wire would be too long for effective operation and would cause, for example, excessive latency, then a warning may be generated to the processor that such wiring, if produced, would be out of specification. In the embodiments, wired connections are understood to mean electrical connections that are established with a component, such as a die of a microprocessor.

[0052] At operation **504**, as shown in FIG. **6B**, a photoresist **620** is deposited over the substrate **140**. The photoresist **620** is sensitive to radiation. A positive photoresist includes portions of the photoresist, which when exposed to radiation, will be respectively soluble to photoresist developer applied to the photoresist after the pattern is written into the photoresist. A negative photoresist includes portions of the photoresist, which when exposed to radiation, will be respectively insoluble to photoresist developer applied to the photoresist after the pattern is written into the photoresist. The chemical composition of the photoresist determines whether the photoresist will be

a positive photoresist or negative photoresist. Examples of photoresists include, but are not limited to, at least one of diazonaphthoquinone, a phenol formaldehyde resin, poly (methyl methacrylate), poly (methyl glutarimide), and SU-8. In this manner, the pattern is created on a surface of the substrate **140** to form the electronic circuitry, e.g. wired connections.

[0053] At operation **505**, as shown in FIG. **6C**, the photoresist **620** is developed to form developed regions **630** and undeveloped regions **650**. The photoresist **620** is developed using the photoresist developer. The photoresist developer is applied to the photoresist **620** to form a pattern corresponding to the altered wired connection **402**. In some embodiments, the developed regions **630** include altered wired connection developed region **602** and misaligned altered wired connection developed region **605**. In other embodiments, there are no misaligned altered wired connection developed regions **605** formed. The altered wired connection developed region **602** corresponds to the altered wired connection **402**.

[0054] At operation **506**, as shown in FIG. **6D**, the developed regions **630** are removed to form an altered wired connection region **640** (e.g., an altered electrical connection template) and a misaligned altered wired connection region **615** (e.g., misaligned regions of the altered electrical connection template). In some embodiments, however, no misaligned altered wired connection regions **615** are formed.

[0055] At operation **507**, the substrate **140** is scanned to obtain a third set of data. At operation **508**, the third set of data is compared to the second set of data. In some embodiments, the third set of data is compared to the second set of data to detect the location and orientation of the altered wired connection region **640**. The location and orientation of the altered wired connection region **640** may be detected using infrared metrology, or by imaging bottom vias. In some embodiments, the third set of data is compared to the second set of data in order to detect misaligned altered wired connection region **615**. The misaligned altered wired connection regions **615** deviates from the altered wired connection **402**. Detecting the misaligned altered wired connection region **615** enables the removal of the photoresist pattern having misaligned altered wired connection region **615** prior to the deposition of the wired connection material.

[0056] At optional operation **509** the undeveloped regions **650** of the photoresist **620** are removed. In embodiments in which misaligned altered wired connection regions **615** are detected, the undeveloped regions **650** of the photoresist **620** are removed to repeat operations **504-508**. Operations **501-508** are repeated until the third set of data does not detect misaligned altered wired connection regions **615**, or the misaligned altered wired connection regions **615**. The removal of the photoresist **620** when misaligned altered wired connection regions **615** are detected enables an increase in defect detection before the more permanent wired connection material is deposited, thus reducing scrap and increasing throughput.

[0057] In some embodiments, hardware architects may determine that slight variations from altered wired connection **402** are allowable. The deviation of the misaligned altered wired connection regions **615** from the altered wired connection **402** may be determined to be within acceptable levels, e.g., a threshold. In those instances, no removal of the undeveloped regions **650** may be required.

[0058] In some embodiments, if the location and orientation of the misaligned altered wired connection regions **615** is outside of the threshold, wired connections cannot be effectively made to the component. A separate warning may be made to the processors that the misaligned altered wired connection regions **615** are out of tolerance and such altered wired connections **402** would be compromised.

[0059] At operation **510**, as shown in FIG. **6E**, a wired connection material **660** is deposited in the altered wired connection region **640** to form an altered wired connection **402**. At operation **511**, as shown in FIG. **6F**, the undeveloped regions **650** are removed to form the altered layout plot **400**.

[0060] FIG. **7** is a flow chart of a method **700** for processing an apparatus in a lithography system. The method **700** is a material subtraction method, i.e., the altered wired connections **402** are

formed by removing a wired connection material outside of the wired connection region. FIGS. 8A-8F are schematic, top views of a substrate **140** during the method **700** for processing an apparatus in a lithography system. In some embodiments, the method **700** for processing the substrate **140** may be performed in a microlithography system, such as photolithography system **100A** or photolithography system **100B**.

[0061] At operation **701** designed data may be obtained. The designed data (e.g., a first set of data) may include coordinate data on a designed component **301** as well as a designed wired connection **302** pattern to the designed component **301** in a designed state. This designed data may be obtained from manufacturing drawings or the memory of the controller, for example. In one embodiment, position data is obtained on the designed component **301** and the designed wired connection **302** to the designed component **301** for a designed state. The designed component **301** may be any type of component that needs wiring attachment, such as, but not limited by, processors. The designed wired connection **302** may be, for example, from the designed component **301** to a fixed perimeter **303**, or to a predetermined location, e.g., the edge **304** of another designed component **301**.

[0062] At operation **702**, as shown in FIG. 8A, a substrate **140** with a plurality of altered components **401** is provided. The substrate **140** is placed on a stage of a microlithography system. As can be understood, the altered component **401** is positioned within a defined field for processing. The altered components **401** may vary slightly from the designed component **301** from various factors, including the ability of locating such designed components **301** accurately during initial processing of the platen (indexing table) before loading of the platen into the microlithography machine. Other factors may also cause the designed component **301** to move, such as handling of the components during microlithography steps.

[0063] At operation **703** the altered components **401** are scanned in-situ to obtain altered component data (e.g., a second set of data). To achieve an altered wired connection **402** to the altered components **401**, either from the fixed perimeter **303** or in between altered components **401**, the “in-situ” positioning of the altered components **401** on the substrate **140** are scanned with at least one scanning device to be able to ascertain the exact positioning of each altered component **401**. In some embodiments, the stage is scanned including the altered component **401** with the at least one scanning device. The substrate **140**, therefore, is moved into a position in the lithography system such that the scanning occurs for processing of coordinate data. As will be understood, scanning of the altered components **401** may also be accomplished by a separate process, if needed, and the data fed to the microlithography machine for use. The altered component data obtained may be stored in the controller. The second set of coordinate data may be compared to the first set of data to determine any offsets that may be present in the altered placement of the components compared to the designed positioning. An offset may be calculated to allow the microlithography machine to understand the exact placement of the components. The wired connection pattern from the designed condition may then be altered using data from the offset calculation. Visual image data may also be used to determine the difference that is needed for the wired connection pattern.

[0064] Computer analysis of the data may be performed to determine the required offset. The computer analysis may not only calculate the offset, but also provide the altered wired connection **402** locations between the altered component **401** and the fixed perimeter **303** or other altered components **401**, therefore speeding processing.

[0065] In embodiments, hardware architects may determine that slight variations from designed placement are allowable. The deviation for the wired connection and the component may be determined to be within acceptable levels. In those instances, no alteration of the wired connection may be required.

[0066] When deviations from these acceptable levels are found, a warning may be generated to the processors to allow for notification that alterations/modification of the wiring are necessary. In other embodiments, if the location and orientation of the components is outside of a maximum threshold, wired connections cannot be effectively made to the component. A separate warning may

be made to the processors that the components are out of tolerance and that even with modification of the wired connection, such wired connections will be compromised.

[0067] As will be understood, designs for wired connections may take into account not only placement accuracy of the starting and ending points of the wired connection, but also wired connection length. If a length of wire would be too long for effective operation and would cause, for example, excessive latency, then a warning may be generated to the processor that such wiring, if produced, would be out of specification. In the embodiments, wired connections are understood to mean electrical connections that are established with a component, such as a die of a microprocessor.

[0068] At operation **704**, as shown in FIG. **8B**, a wired connection material **860** is deposited over the substrate **140**. The wired connection material may include aluminum, copper, tungsten, or combinations thereof.

[0069] At operation **705**, as shown in FIG. **8C**, a photoresist **820** is deposited over the substrate **140**. The photoresist **820** is sensitive to radiation, and may be a positive photoresist or a negative photoresist. In this manner, the pattern is created on a surface of the substrate **140** to form the electronic circuitry, e.g. wired connections.

[0070] At operation **706**, as shown in FIG. **8D**, the photoresist **820** is developed to form developed regions **830** and undeveloped regions **850**. The photoresist **820** is developed using the photoresist developer. The photoresist developer is applied to the photoresist **820** to form a pattern corresponding to the altered wired connection **402**. In some embodiments, the undeveloped regions **850** include altered wired connection undeveloped regions **802** and misaligned altered wired connection undeveloped region **805**. In other embodiments, there are no misaligned altered wired connection undeveloped regions **805** formed. The altered wired connection undeveloped region **802** corresponds to the altered wired connection **402**.

[0071] At operation **707**, as shown in FIG. **8E**, the developed regions **830** are removed to form an altered wired connection region **840** (an altered electrical connection template) and a misaligned altered wired connection region **815** (e.g., misaligned regions of the altered electrical connection template). The removal of the developed regions **830** exposes the wired connection material **860**. In some embodiments, however, no misaligned altered wired connection regions **815** are formed.

[0072] At operation **708**, the substrate **140** is scanned to obtain a third set of data. At operation **709** the third set of data is compared to the second set of data. In some embodiments, the third set of data is compared to the second set of data in order to detect the location and orientation of the altered wired connection region **840**. The location and orientation of the altered components **401** and the altered wired connection region **840** may be detected using infrared metrology, or by imaging bottom vias. In some embodiments, the third set of data is compared to the second data set in order to detect misaligned altered wired connection region **815**. The misaligned altered wired connection regions **815** deviates from the altered wired connection **402**. Detecting the misaligned altered wired connection region **815** enables the removal of the photoresist pattern having misaligned altered wired connection region **815** prior to the removal of the wired connection material **860** to form the altered wired connections.

[0073] In some embodiments, hardware architects may determine that slight variations from altered wired connection **402** are allowable. The deviation of the misaligned altered wired connection regions **815** from the altered wired connection **402** may be determined to be within acceptable levels, e.g., a threshold. In those instances, no removal of the undeveloped regions **650** may be required.

[0074] In other embodiments, if the location and orientation of the misaligned altered wired connection regions **815** is outside of the threshold, wired connections cannot be effectively made to the component. A separate warning may be made to the processors that the misaligned altered wired connection regions **815** are out of tolerance and such altered wired connections **402** would be compromised.

[0075] At optional operation **710**, the undeveloped regions **850** of the photoresist **820** are removed. In other words, at least a portion of the altered electrical connection template is removed. In embodiments in which misaligned altered wired connection regions **815** are detected, the undeveloped regions **850** of the photoresist **820** are removed to repeat operations **705-709**. Operations **705-709** are repeated until the third set of data does not detect misaligned altered wired connection regions **815**. The removal of the photoresist **820** when misaligned altered wired connection regions **815** are detected enables an increase in defect detection before the more permanent wired connection material is removed, thus reducing scrap and increasing throughput.

[0076] At operation **711**, as shown in FIG. **8F**, the undeveloped regions **850** and the exposed wired connection material **860** are removed to form the altered layout plot **400**.

[0077] In summation, the method is disclosed. The method includes determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate. The method includes depositing a photoresist over the substrate and developing the photoresist to form one or more developed regions and one or more undeveloped regions. One of the one or more developed regions or the one or more undeveloped regions of the photoresist are removed to form an altered electrical connection template. The altered component and the altered electrical connection template are scanned to identify one or more misaligned regions of the altered electrical connection template. At least a portion of the altered electrical connections template is removed. The disclosure provides for measuring the location and orientation of the adaptively generated wiring to detect defects in the adaptively generated wiring prior to the deposition of the wiring material, thereby improving throughput in fabrication.

[0078] While embodiments have been described herein, those skilled in the art, having benefit of this disclosure will appreciate that other embodiments are envisioned that do not depart from the inventive scope of the present application. Accordingly, the scope of the present claims or any subsequent related claims shall not be unduly limited by the description of the embodiments described herein.

Claims

1. A method for processing an apparatus, comprising: determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate; depositing a photoresist over the substrate; developing the photoresist to form one or more developed regions and one or more undeveloped regions; removing one of the one or more developed regions or the one or more undeveloped regions of the photoresist to form an altered electrical connection template; scanning the altered component and the altered electrical connection template to identify one or more misaligned regions of the altered electrical connection template; and removing at least a portion of the altered electrical connections template.
2. The method according to claim 1, further comprising, based on determining that no misaligned regions are identified, forming the one or more altered electrical connections on the altered electrical connection template.
3. The method according to claim 2, wherein the one or more developed regions correspond to the one or more altered electrical connections, and the forming of the one or more altered electrical connections comprise: depositing an electrical connection material in the altered electrical connection template; and removing at least a portion of the altered electrical connections template to form an altered layout plot.
4. The method according to claim 2, wherein the one or more undeveloped regions correspond to the one or more altered electrical connections, and the forming of the one or more altered electrical connections comprises: depositing an electrical connection material over the substrate prior to depositing the photoresist; and removing at least a portion of the altered electrical connections

template and an exposed portion of the electrical connection material to form an altered layout plot, wherein removing the one or more developed regions of the photoresist to form an altered electrical connection template exposes a portion of the electrical connection material.

5. The method according to claim 1, further comprising: based on determining that the offset is less than a threshold, setting the offset to zero.

6. The method according to claim 1, further comprising: based on determining that the offset is greater than a threshold, generating a warning that the threshold has been exceeded.

7. The method according to claim 1, wherein the method is performed in-situ in a maskless microlithography system.

8. The method according to claim 1, further comprising providing the one or more altered electrical connections between the altered component and a second altered component.

9. The method according to claim 1, further comprising providing the one or more altered electrical connections between the altered component and a fixed perimeter of the substrate.

10. A method for processing an apparatus, comprising: determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate; performing a photoresist development process, comprising: depositing a photoresist over the substrate; developing the photoresist to form one or more developed regions and one or more undeveloped regions; removing the one or more developed regions or the one or more undeveloped regions of the photoresist to form an altered electrical connection template; scanning the altered component and the altered electrical connection template to identify one or more misaligned regions of the altered electrical connection template; removing at least a portion of the altered electrical connection template upon identifying the one or more misaligned regions; and repeating the photoresist development process upon identifying the one or more misaligned regions; and forming the one or more altered electrical connections.

11. The method according to claim 10, wherein the one or more developed regions correspond to the one or more altered electrical connections, the forming of the one or more altered electrical connections comprises: depositing an electrical connection material in the altered electrical connection template; and removing at least a portion of the altered electrical connections template to form an altered layout plot.

12. The method according to claim 10, wherein the one or more undeveloped regions correspond to the one or more altered electrical connections, the forming of the one or more altered electrical connections comprises: depositing an electrical connection material over the substrate prior to depositing the photoresist; and removing at least a portion of the altered electrical connections template and an exposed portion of the electrical connection material to form an altered layout plot, wherein removing the one or more developed regions of the photoresist to form the altered electrical connections template exposes a portion of the electrical connection material.

13. The method according to claim 10, further comprising: further comprising providing the one or more altered electrical connections between the altered component and a second altered component.

14. The method according to claim 10, further comprising: comparing a first set of data of a designed component to a second set of data of the altered component; and comparing the offset to a threshold, wherein: when the offset is less than the threshold, setting the offset to zero; and when the offset is greater than the threshold, generating a warning to a user that the threshold has been exceeded.

15. The method according to claim 10, wherein the one or more altered electrical connections is based upon the offset.

16. A method for processing an apparatus, comprising: determining one or more altered electrical connections based on an offset between a designed component and an altered component disposed on a substrate; performing a photoresist development process, comprising: depositing a photoresist over the substrate; developing the photoresist to form one or more developed regions and one or

more undeveloped regions; removing the one or more developed regions or the one or more undeveloped regions of the photoresist to form an altered electrical connection template; scanning the altered component and the altered electrical connection template to identify one or more misaligned regions of the altered electrical connection template; removing at least a portion of the altered electrical connection template upon identifying the one or more misaligned regions; and repeating the photoresist development process upon identifying one or more misaligned regions; depositing an electrical connection material in the altered electrical connection template; and removing at least a portion of the altered electrical connections template to form an altered layout plot.

17. The method according to claim 16, wherein the method is performed in-situ in a maskless microlithography system.

18. The method according to claim 16, further comprising: further comprising providing the one or more altered electrical connections between the altered component and a second altered component.

19. The method according to claim 16, further comprising: comparing a first set of data of a designed component to a second set of data of the altered component; and comparing the offset to a threshold, wherein: when the offset is less than the threshold, setting the offset to zero; and when the offset is greater than the threshold, generating a warning to a user that the threshold has been exceeded.

20. The method according to claim 16, wherein the one or more altered electrical connections is based on the offset.
